

Hanxi “Gary” Chen

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EDUCATION

Cornell University

Ph.D. in Computer Science, Programming Languages
Advisors: Prof. Alexandra Silva, Prof. Andrew C. Myers

Ithaca, NY
2025 – Present

University of Pennsylvania, Penn Engineering & The Wharton School

M.S.E. in Computer and Information Science
B.S.E. in Computer Science, Mathematics minor
B.S. in Economics

Philadelphia, PA
May 2025

GPA: 3.97/4.00, *Summa Cum Laude*

Advisors: Prof. Steve Zdancewic, Prof. Sanjeev Khanna

PUBLICATIONS

Refereed

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. [Vellvm: Formalizing the Informal LLVM \(An Experience Report\)](#). NASA Formal Methods: 17th International Symposium (NFM), 2025.

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. [Vellvm: Formalizing the Informal](#). The Eleventh International Workshop on Coq for Programming Languages (CoqPL), 2025. Presented with Calvin Beck; also given at PLClub, University of Pennsylvania, Jan 2025.

Calvin Beck, Irene Yoon, **Hanxi Chen**, Yannick Zakowski, Steve Zdancewic. [A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer–Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction](#). Proceedings of the ACM on Programming Languages, 8 (ICFP), 2024. [MIT PL Review 2025]

Preprints & Manuscripts

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. [Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary](#). arXiv, 2026. Presented at the Cornell PL Seminar (PLDG), Sep 2026.

Silei Ren, Suraaj Kanniwadi, **Hanxi Chen**, Andrew C. Myers. Type-Confusion Security. 2026.

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. GenLLVM: Randomized Differential Testing for LLVM IR. 2025.

Theses

Hanxi Chen. Knapsack Problems with Precedence Constraints. Wharton Undergraduate Thesis, 2024. Supervised by Prof. Sanjeev Khanna.

Hanxi Chen. Random Differential Testing at the LLVM IR Level. CIS Undergraduate Thesis, 2024. Reviewed by Prof. Steve Zdancewic (advisor) and Prof. Benjamin C. Pierce.

RESEARCH EXPERIENCE

Research Assistant, *Cornell PL Group*, advised by Prof. Alexandra Silva and Prof. Andrew C. Myers

Sep 2025 – Present

Cornell University, Computer Science

- Working toward end-to-end security for probabilistic and concurrent systems, from adversary models to semantics-preserving compilation.
- Developed Oblivious Probabilistic Outcome Logic (opOL), a logic for probabilistic nondeterministic programs with oblivious adversaries; used schedule consumption and probabilistic independence to verify randomized caching and leader-election protocols.
- Mechanized opOL in Lean 4, formalizing monadic semantics, probability-space constructions, probabilistic assertions, semantic and syntactic outcome triples, soundness proofs, and verified case studies.
- Led the Lean mechanization of Type-Confusion Security, proving type safety for the static and dynamic information-flow checks that protect decentralized systems whose untrusted components can misrepresent dynamic types.

Research Mentor, *Cornell Math+AI*, advised by Prof. Daniel Halpern-Leistner

Jan 2026 – Present

Cornell University, Mathematics

- Mentored the Jan–Aug 2026 iteration of an autoformalization project, producing a proof-quality evaluation platform and LaTeX-to-Lean translation tools with interactive dependency graphs; now also directing a newer project on large language models for program reasoning.

Research Assistant, *PLClub*, advised by Prof. Steve Zdancewic

Jun 2021 – Aug 2025

University of Pennsylvania, Computer and Information Science

- Contributed to the ICFP 2024 Vellvm memory-model paper in Prof. Steve Zdancewic's group, helping design a low-level memory model for LLVM optimization reasoning, verifying serialization lemmas in Rocq, and implementing tests for the memory model.
- Built Rocq/OCaml validation tools for testing Vellvm's semantic faithfulness against LLVM implementations, including QuickChick generators for UB-free LLVM programs, interpreter-based traces, and a differential-testing framework that found novel LLVM bugs.
- Developed Rocq infrastructure for permutation reasoning and list AC unification, then applied it to formalize first-order classical linear logic with cut elimination and a linear-resource calculus for asynchronous queries.
- Contributed to DeepSpec Software Foundations material presented at the Oregon Programming Languages Summer School 2024.

Research Assistant, *Wharton Research Scholars*, advised by Prof. Sanjeev Khanna
University of Pennsylvania, The Wharton School

Jan 2023 – Dec 2024

- Studied precedence-constrained knapsack problems, which are strongly NP-hard in general, and designed pseudo-polynomial algorithms for structured poset classes including unions of three paths and one-directional bipartite posets with consecutive children.
- Presented the work in the Wharton Research Scholars seminar and wrote an undergraduate thesis.

HONORS & AWARDS

MCIT TA Award	2024	CIS Senior Thesis	2024
CIS Faculty Appreciation Award	2024	Wharton Research Scholars	2024

Honor societies: Eta Kappa Nu, Tau Beta Pi, Beta Gamma Sigma

TEACHING EXPERIENCE

Graduate Teaching Assistant, *Formal Verification (CS 4160 / CS 5160)*

Cornell University. Instructor: Prof. Michael Clarkson. Terms: Spring 2026.

Graduate Teaching Assistant, *Programming Languages and Logics (CS 4110 / CS 5110)*

Cornell University. Instructor: Prof. Nate Foster. Terms: Fall 2025.

Teaching Assistant, *Compilers and Interpreters (CIS 5521)*

University of Pennsylvania. Instructor: Prof. Steve Zdancewic. Terms: Spring 2025.

Teaching Assistant, *Advanced Programming (CIS 5520)*

University of Pennsylvania. Instructor: Prof. Stephanie Weirich. Terms: Fall 2024.

Taught four lectures and mentored four final-project groups.

Head Teaching Assistant, *Analysis of Algorithms (CIS 5020 / CIT 5960 / CIS 3200)*

University of Pennsylvania. Instructor: Prof. Sanjeev Khanna. Terms: Spring 2023, Summer 2023, Fall 2023, Summer 2024.

Teaching Assistant for the course in Summer 2022 and Fall 2022. Led TA staff and student support across six terms; received the 2024 MCIT TA Award.

Head Teaching Assistant, *Theory of Networks (NETS 3120)*

University of Pennsylvania. Instructor: Prof. Victor Preciado. Terms: Spring 2024.

Teaching Assistant, *Software Foundations (CIS 5000)*

University of Pennsylvania. Instructor: Prof. Benjamin C. Pierce. Terms: Fall 2023.

Taught one lecture on maps in Rocq mechanization.

STEM Tutor, *The Weingarten Center*

University of Pennsylvania. Terms: Spring 2021, Summer 2021, Fall 2021, Spring 2022, Summer 2022.

Tutored core CS and mathematics courses.

ACADEMIC SERVICE

Seminar Organization: Co-Czar, Cornell PL Seminar (PLDG), 2026

Artifact Evaluation Committee: ATVA 2026, PLDI 2026, POPL 2026

Student Volunteer: POPL 2025, SPLASH 2024, OPLSS 2024

SELECTED TECHNICAL PROJECTS

RISC-V Pipelined Processor – *SystemVerilog, Python, FPGA*

Feb 2025 – May 2025

University of Pennsylvania, *CIS 5710 Computer Organization and Design*

- Designed a RISC-V pipelined processor with AXI-Lite caching, multi-cycle units, branch prediction, and memory-mapped I/O.
- Built verification testbenches, optimized LUT usage and timing behavior, and synthesized the design to FPGA to run compiled programs.

Oat Compiler – *OCaml, C, LLVM IR, x86, Clang, Dune*

Feb 2024 – May 2024

University of Pennsylvania, *CIS 7000 (now CIS 5521) Compilers and Interpreters*

- Implemented a compiler for Oat, a C-like language with pointers and structured data, including lexing, parsing, type checking, frontend lowering, and backend code generation to LLVM IR and x86.
- Added static analyses and optimizations including mem2reg and dead-code elimination, validated against thousands of test cases.

SQL Interpreter – *Haskell, MySQL, Cabal*

Sep 2023 – Dec 2023

University of Pennsylvania, *CIS 5520 Advanced Programming*

- Led development of a Haskell library for parsing, interpreting, optimizing, and pretty-printing SQL, using GADTs, monad transformers, applicative parsers, and property-based testing.
- Designed typed abstract syntax and execution interfaces for local query evaluation.